# Awni Altabaa

PhD Candidate · Statistics & Data Science, Yale University

awni.altabaa@yale.edu
awni.xyz
Google Scholar
github.com/awni00
linkedin.com/in/awni-altabaa

I work on the foundations of artificial intelligence, developing new architectures, algorithms, and theoretical frameworks to identify simple, unifying principles that explain how intelligent systems learn.

## RESEARCH

**Architectural Inductive Biases**
What mechanisms an architecture needs for models to learn efficiently and form robust abstractions.
[3] [5] [7] [8] [9] [10]

**Reasoning in Transformers & LLMs**
How models learn, represent, and execute multi-step reasoning, and how to improve out-of-distribution generalization.
[1] [2] [3] [4]

**Theoretical Foundations**
Formal accounts of how architecture, learning algorithm, and problem structure determine what models can learn.
[1] [4] [6] [7] [11]

## EDUCATION

**Ph.D. in Statistics & Data Science**, Yale University, New Haven, CT — 2022 – 2027 (expected)
Advisor: John Lafferty

**Master's in Statistics**, Yale University, New Haven, CT — 2022 – 2024

**Bachelor's in Applied Mathematics**, Queen's University, Kingston, ON — 2018 – 2022

## EXPERIENCE

**Research Intern**, Microsoft Research, AI Interaction & Learning group — Summer 2026
Paper: *Putting Transformer Theories to the Test* [1]

**Quantitative Researcher Intern**, Citadel Securities — Summer 2025
Systematic Equities / FICC. End-to-end time-series forecasting alpha research: data collection, feature generation with an LLM agent system (RAG), and alpha modeling.

## PUBLICATIONS

1. **Putting Transformer Theories to the Test**
   **A. Altabaa**, T. Schnabel, and K. Tomlinson
   *arXiv preprint*, 2026 awni.xyz/blog/transformer-theories arXiv:2610.32104

2. **Uncovering Latent Reasoning Strategies in Language Models**
   **A. Altabaa** and J. Lafferty
   *arXiv preprint*, 2026 awni.xyz/blog/latent-strategies-in-lms arXiv:2607.17674

3. **Unlocking Out-of-Distribution Generalization in Transformers via Recursive Latent Space Reasoning**
   **A. Altabaa**, S. Chen, J. Lafferty, and Z. Yang
   *arXiv preprint*, 2025 awni.xyz/blog/recursive-latent-reasoning arXiv:2510.14095

4. **CoT Information: Improved Sample Complexity under Chain-of-Thought Supervision**
   **A. Altabaa**, O. Montasser, and J. Lafferty
   *Neural Information Processing Systems (NeurIPS)*, 2025 · Spotlight awni.xyz/blog/cot-info arXiv:2505.15927

5. **Disentangling and Integrating Relational and Sensory Information in Transformer Architectures**
   **A. Altabaa** and J. Lafferty
   *International Conference on Machine Learning (ICML)*, 2025 awni.xyz/blog/dual-attention arXiv:2405.16727

6. **On the Role of Information Structure in Reinforcement Learning for Partially-Observable Sequential Teams and Games**
   **A. Altabaa** and Z. Yang
   *Neural Information Processing Systems (NeurIPS)*, 2024 awni.xyz/blog/information-structure-rl arXiv:2403.00993

7. **Approximation of Relation Functions and Attention Mechanisms**
   **A. Altabaa** and J. Lafferty

*Information Theory, Probability and Statistical Learning: A Festschrift in Honor of Andrew Barron*, 2024
awni.xyz/blog/relation-function-approximation arXiv:2402.08856

8 **Learning Hierarchical Relational Representations through Relational Convolutions**
**A. Altabaa** and J. Lafferty
*Transactions on Machine Learning Research (TMLR)*, 2024 awni.xyz/blog/relational-convolutions arXiv:2310.03240

9 **The Relational Bottleneck as an Inductive Bias for Efficient Abstraction**
T. W. Webb, S. M. Frankland, **A. Altabaa**, S. Segert, K. Krishnamurthy, D. Campbell, J. Russin, T. Giallanza, *et al.*
*Trends in Cognitive Sciences (TICS)*, 2024 awni.xyz/blog/relational-bottleneck arXiv:2309.06629

10 **Abstractors and Relational Cross-Attention: An Inductive Bias for Explicit Relational Reasoning in Transformers**
**A. Altabaa**, T. Webb, J. Cohen, and J. Lafferty
*International Conference on Learning Representations (ICLR)*, 2024 awni.xyz/blog/abstractor arXiv:2304.00195

11 **Decentralized Multi-Agent Reinforcement Learning for Continuous-Space Stochastic Games**
**A. Altabaa**, B. Yongacoglu, and S. Yüksel
*2023 IEEE American Control Conference (ACC)*, 2023 awni.xyz/blog/decentralized-marl-continuous-spaces arXiv:2303.13539

## HONORS

Rising Stars in Statistics & Data Science, Cornell University — Sep 2026
NeurIPS Spotlight, *CoT Information* — 2025
NSF AI Institute for Artificial and Natural Intelligence (ARNI), Fellow/Trainee

## TALKS & PRESENTATIONS

**Rising Stars in Statistics & Data Science, Cornell University** Poster — Sep 2026

**Formal Languages and Neural Networks (FLaNN) Seminar** Invited talk — Feb 2026
*Unlocking Out-of-Distribution Generalization in Transformers via Recursive Latent Space Reasoning* [3]

**NeurIPS 2025, San Diego** Spotlight — Dec 2025
*CoT Information: Improved Sample Complexity under Chain-of-Thought Supervision* [4]

**Formal Languages and Neural Networks (FLaNN) Seminar** Invited talk — Oct 2025
*CoT Information: Improved Sample Complexity under Chain-of-Thought Supervision* [4]

**Deep Learning: Classics and Trends (DLCT), ML Collective** Invited talk — May 2025
*Disentangling and Integrating Relational and Sensory Information in Transformer Architectures* [5]

**Two Sigma PhD Symposium** Invited talk — Jun 2024

**New England Statistics Symposium (NESS 2024)** Presentation — May 2024
*On the Role of Information Structure in Reinforcement Learning for Partially-Observable Sequential Teams and Games* [6]

**Yale Wu Tsai Institute Workshop on Abstraction and Relational Reasoning** Presentation — Nov 2023
*Learning Hierarchical Relational Representations through Relational Convolutions* [8]

**American Control Conference (ACC 2023), San Diego** Presentation — Jun 2023
*Decentralized Multi-Agent Reinforcement Learning for Continuous-Space Stochastic Games* [11]

## TEACHING EXPERIENCE

**Introductory Machine Learning** (S&DS 2650), Yale University — Spring 2026
**Intermediate Machine Learning** (S&DS 3650), Yale University — Fall 2025
**Theory of Reinforcement Learning** (S&DS 685), Yale University — Spring 2025
**Introductory Machine Learning** (S&DS 265), Yale University — Fall 2023

## SERVICE

**Reviewer:** NeurIPS (2024, 2025, 2026), ICLR (2025, 2026), ICML (2025, 2026; Gold Reviewer Award, 2026), Journal of Data Science